VISITOR ENTRY TRACKING SYSTEM USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

AKASH M

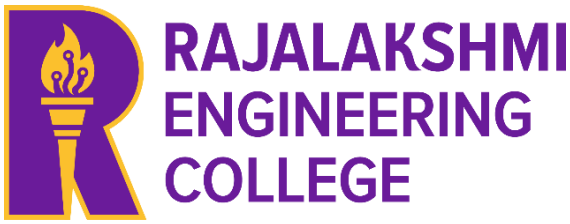
(240701025)

in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "Visitor Entry Tracking System using Zoho Creator" is the bonafide work of AKASH M (240701025),who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, AKASH M (240701025), hereby declare that the project report titled “Visitor Entry Tracking System using Zoho Creator”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, dashboard page, Deluge scripts, workflows and Blueprint — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature, code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

Date: 10.11.2026

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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

AKASH M

ABSTRACT

Educational institutions, offices and other controlled premises receive a steady flow of visitors every day — parents, guests, vendors, interview candidates and official callers. In many places the gate or reception still records these visits in a paper register: the visitor's name, phone number and purpose are written by hand, the host is phoned for confirmation, and the exit time is often not recorded at all. This makes it hard to know who is inside the premises at a given moment, allows illegible or incomplete entries, gives a visitor no unique identity, and leaves department heads unaware that a visitor has arrived to meet them.

This project presents a Visitor Entry Tracking System developed using Zoho Creator. It consists of four forms — Visitor Registration, Visitor Log, Visitor History and Department Master — and seven reports (four list reports and three kanban boards). The Visitor Log and Visitor History forms are linked to the registration record through lookup fields. Three form workflows written in Deluge generate a unique, locked Visitor ID when the registration form opens, reject an exit time earlier than the entry time, and email the department head when a visitor with the status "Inside" is registered. A scheduled workflow compiles a daily occupant summary of all visitors currently inside and emails it as an HTML table to the security team. A Blueprint, "Visitor Entry and Checkout Flow", guides each registration through eight stages and eleven transitions. A dashboard page shows record counts, action counts, two bar charts (distribution by department and by purpose of visit), quick-link buttons and an embedded Visitor Log report. Access is controlled through a CEO role and eleven permission profiles, including Receptionist, Security Officer, Department Head, System Administrator, Host Employee and Visitor.

The application replaces the paper register with one structured system, gives every visit a unique ID, reduces manual communication, enforces basic data checks, and gives security staff and management a live view of visitors on the premises. Test cases, deployment details, a candid defect log and the measured outcomes are presented in Chapters 6 to 8.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Visitor Entry Tracking System using Zoho Creator
Application Domain	Facility and campus security – visitor management
Student Name & Register No.	AKASH M – 240701025
Zoho Creator Edition	Free
Application Name (Link Name)	Visitor Entry Tracking System (visitor-entry-tracking-system)
Published / Live URL	https://creator.zoho.in/userhome/akash.m.2024.cse_rajalaks367
Demo Credentials (per role)	Admin (application owner): akash.m.2024.cse_rajalaks367. Other roles are configured as permission profiles (Receptionist, Security Officer, Department Head, System Administrator, Host Employee, Visitor).
Number of Forms / Subforms	4 / 0
Number of Reports (by type)	List: 4, Kanban: 3 (total 7)
Number of Pages / Dashboards	1 (DashBoard)
Number of Workflows	Form: 3, Schedule: 1, Approval: 0, Blueprint: 1, Batch: 0
Deluge Functions / Custom Actions	4 scripts (DS-01 to DS-04); custom functions: 0; custom actions: 0
Integrations	None (e-mail sent through the Deluge sendmail task)
Roles & Profiles Defined	Role: CEO. Permission profiles (11): Administrator, Developer, Read, Write, Receptionist, Security Officer, Department Head, System Administrator, Host Employee (portal), Visitor (portal), Customer (portal)
Environments Used	Development-Production
Date of Production Deployment	05.11.2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CEO	Chief Executive Officer
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR	Functional Requirement
HoD	Head of Department
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface

CHAPTER 1
INTRODUCTION

1.1 Background

Every campus, office and factory has to know who is on its premises. Visitors arrive to meet staff, attend interviews, deliver goods or visit students, and each visit involves the same basic steps: identifying the visitor, recording the purpose and the person to be met, informing the host, noting the entry time and later noting the exit. Security staff, receptionists, department heads and management all depend on this information.

This project develops a Visitor Entry Tracking System using Zoho Creator. It centralises visitor registrations, entry and exit logs, visit history and department contact details in four linked forms, automates identification and notification through Deluge workflows, guides each visit through a Blueprint, and controls who can see or change what through permission profiles.

1.2 Problem Statement

Recording visitors manually is slow and unreliable. Entries are handwritten and may be unreadable or incomplete, a visitor has no unique identity that can be quoted later, exit times are frequently missed, hosts are not told when their visitor arrives, and nobody can say with confidence how many visitors are inside the building at a given time. A centralised application is needed to record each visit once, validate it, notify the right person and show the current position at a glance.

The main pain points are:

- Visitor details are written in a paper register, so they are hard to search and easy to alter.
- There is no unique Visitor ID, so a visit cannot be traced reliably across entry, exit and history.
- Exit time can be missing or earlier than the entry time because nothing checks it.
- Department heads are not informed automatically when a visitor arrives to meet someone in their department.
- Security staff cannot quickly list the visitors who are still inside the premises, which matters during an emergency or at closing time.
- Anyone who can pick up the register can read or modify every record.

1.3 Objectives

- To design forms that capture visitor registration, entry/exit action logs, visit history events and department master data.
- To automate Visitor ID generation, exit-time validation, department-head notification, a daily occupant summary and visit-stage tracking using Deluge workflows, a scheduled workflow and a Blueprint.
- To provide reports, kanban boards, a dashboard and role-based access for the Receptionist, Security Officer, Department Head, System Administrator, Host Employee and Visitor.
- To use the e-mail capability of Zoho Creator (the Deluge sendmail task) for alerting department heads and the security team.

1.4 Scope of the Project

In Scope: Four forms with their list reports and kanban boards (seven reports), three form workflows, one scheduled workflow, one Blueprint with eight stages, one dashboard page with record counts, action counts, two bar charts, quick-link buttons and an embedded report, a CEO role, and eleven permission profiles. The application is used through the Zoho Creator web interface.

Out of Scope: Barcode/QR badge printing, photo or ID-proof capture, visitor blacklisting, integration with access-control hardware, SMS/WhatsApp alerts, an approval process, AI features and a custom mobile application.

1.5 Justification for a Low-Code Approach

Low-code platforms simplify application development through visual tools and a small amount of scripting [1]. Zoho Creator allows forms, reports, workflows and permission profiles to be configured visually, while Deluge handles the few places where custom logic is needed — here, generating an ID, comparing two date-times and composing e-mails [2], [3], [6]. Hosting, the database, user management and e-mail delivery are provided by the platform, which shortens development time and removes the need to maintain servers.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 covers requirements, Chapter 4 the system design and Chapter 5 the implementation. Chapter 6 discusses testing, Chapter 7 deployment and security, Chapter 8 results, and Chapter 9 the conclusion and future

enhancements. The appendices contain the feature checklist, course-outcome mapping, complete Deluge source code, user manual and the evaluation sheets.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

At present the visitor process at many gates and reception desks follows these steps. (1) The visitor arrives and is stopped by security. (2) Security writes the name, phone number and purpose in a paper register. (3) Security phones the staff member to be met to confirm. (4) The visitor is allowed in and the entry time is noted by hand. (5) The exit time is written when the visitor leaves — or is forgotten. (6) When a record is needed later, the register is searched page by page.

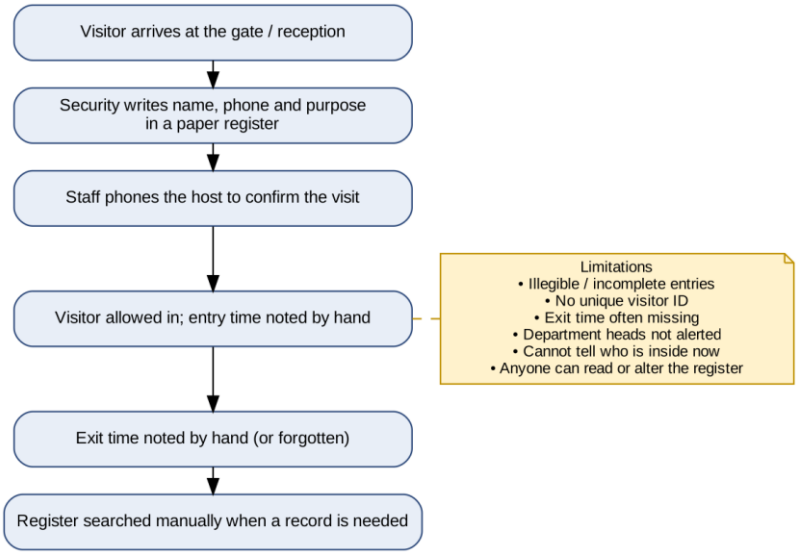


Figure 2.1: As-Is Process Flow of the Existing Visitor Management Process

2.2 Limitations of the Existing System

- Entries are handwritten, so they may be illegible, incomplete or inconsistent, and the register is hard to search.
- There is no unique identifier for a visit, so the same visitor cannot be traced across entry, exit and history.
- There is no validation, so an exit time earlier than the entry time or a missing exit time goes unnoticed.
- Hosts and department heads are informed manually by phone, or not at all.
- There is no real-time list of people inside the premises, which is a safety risk.
- There is no access control: anyone with the register can read or change any entry.
- Summaries such as visitors per department or per purpose must be counted by hand.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Separate design and coding of the database, back end and front end.	Forms, reports, workflows and the dashboard are configured visually, so the application is built in much less time.
Skills required	Programming, database and web development skills.	Understanding of the visitor process and Zoho Creator configuration; Deluge only for the four scripts needed.
Hosting & infrastructure	Servers or cloud services must be arranged and maintained.	Hosted on Zoho's cloud as part of the platform.
Mobile application	A separate mobile app must be developed.	The application can be opened in the Zoho Creator mobile app without separate development.
Security & access control	Authentication, roles and permissions must be coded.	Built-in users, roles and permission profiles with module- and report-level permissions.
Maintenance & change	Every change needs code changes, testing and redeployment.	Changes are made in the builder and published.
Customisation limits	Practically unlimited customisation.	Limited to platform features; complex needs may require pro-code extensions.
Cost	Higher development, hosting and maintenance cost.	Subscription cost with lower development and maintenance effort.

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Zoho Corporation [2]	Zoho Creator Help Documentation	Describes forms, reports, pages, workflows, Blueprints, schedules and user/permission management.	Primary reference for configuring every component of the application.
2	Zoho Corporation [3]	Deluge Scripting Reference	Describes the Deluge language: variables, conditions, record fetch, loops and the sendmail task.	Basis for the four Deluge scripts (DS-01 to DS-04).
3	A. Sahay et al., 2020 [1]	Supporting the understanding and comparison of low-code development platforms	Compares several low-code platforms using a conceptual framework to match platforms to requirements.	Supports the choice of a low-code platform for a form-and-workflow application.
4	D. Di Ruscio et al., 2022 [4]	Low-code development and model-driven engineering: Two sides of the same coin?	Examines the relationship between low-code development and model-driven engineering.	Explains why the data model (forms and lookups) is the starting point of the application.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
5	S. Käss et al., 2023 [5]	Practitioners' perceptions on the adoption of low code development platforms	Reports practitioners' views on benefits and challenges of adopting low-code platforms.	Context for the benefits and limitations discussed in Sections 1.5 and 8.4.

2.5 Emerging Trends in Low-Code Development

Trends such as citizen development, hyper-automation, AI-assisted application building and composable applications are shaping how business applications are delivered [1], [5]. This project follows the citizen-developer pattern: the data model and most automation are built visually, and Deluge is used only for small, well-defined pieces of logic.

2.6 Proposed System

The proposed system records each visitor once in a Visitor Registration form with an automatically generated unique ID, links every action and history event to that ID through lookup fields, validates exit times, notifies the department head on entry, reports the visitors currently inside every day, tracks each visit through a Blueprint, and restricts every user to the actions their role requires.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role / Profile	Description	Key Responsibilities in the Application
Administrator / System Administrator	Owner and developer of the application; configures departments and system data.	Create and configure the application, add users, assign roles and profiles, maintain department codes and active status, view and edit all data.
CEO	Top-level role in the role hierarchy.	Oversees visitor activity; can access the data of all other users.
Receptionist	Registers visitors and verifies identity on arrival.	Create and edit Visitor Registration and Visitor Log records; view Department Master and Visitor History.
Security Officer	Monitors who is currently inside the facility.	Create and edit Visitor Log records; view registrations and history; receives the daily occupant summary.
Department Head	Manages department details and is notified of visitors.	Maintain Department Master (name, code, email, phone, location, active status); view registrations, logs and history; receives entry e-mails.
Host Employee (portal)	Staff member who receives alerts when a scheduled visitor arrives.	Create log entries; view registrations and history.
Visitor (portal)	Provides personal details and purpose of visit during registration.	Create a visitor registration; view log and history reports.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall register a visitor's name, mobile number, purpose of visit, person to meet, department, entry date and time, status and exit date and time.	Receptionist, Visitor	High
FR-02	The system shall generate a unique Visitor ID (format VIS-####-yyyyMMddHHmmss) when the registration form opens and lock the field against editing.	System	High
FR-03	The system shall reject a registration update in which the exit date and time is earlier than the entry date and time.	System	High
FR-04	The system shall record every Check In, Checkout and Update action in the Visitor Log with date, time, previous status, new status and remarks.	Receptionist, Security Officer, Host Employee	High

FR ID	Requirement	Role(s)	Priority
FR-05	The system shall record visit history events (Entry, Out, Registration) with details, recorder and a confirmation flag.	System Administrator	Medium
FR-06	The system shall maintain a Department Master with head of department, e-mail, phone, location and active status.	Department Head, System Administrator	High
FR-07	The system shall e-mail the department head when a visitor is registered with the status “Inside” and the department has a contact e-mail.	System	High
FR-08	The system shall run a scheduled workflow that e-mails the security team a table of all visitors currently inside, or a “no occupants” message.	System	Medium
FR-09	The system shall display the records of each form in a list report, and provide kanban boards grouped by purpose of visit, action taken and event type.	All roles	High
FR-10	The system shall highlight values in list reports using conditional formatting.	All roles	Low
FR-11	The system shall track each registration through the Blueprint stages Visitor Registration Initiated, Visitor Checked In, Visitor On Premises, Visitor Extended Stay, Visitor Checked Out, Visitor Re-Entry, Registration Rejected and Visitor Record Archived.	Receptionist, Security Officer	Medium
FR-12	The system shall provide a dashboard with record counts, action counts, department and purpose bar charts, quick links and an embedded Visitor Log report.	All roles	Medium
FR-13	The system shall restrict access using a role hierarchy and permission profiles with module-level and report-level permissions.	Administrator	High
FR-14	The system shall link Visitor Log and Visitor History records to a registration through a lookup showing the Visitor ID.	System	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and the dashboard shall load promptly (target: within 5 seconds) for the expected daily visitor volume.
NFR-02	Security	Users shall access only the modules, reports and actions permitted by their permission profile.
NFR-03	Usability	The application shall be usable through a standard web browser with simple form-based data entry and one-click quick links.
NFR-04	Availability	The application shall be hosted on the Zoho cloud and available whenever an internet connection is available.
NFR-05	Maintainability	Forms, workflows and permissions shall be modifiable through the visual builder with minimal scripting.

NFR ID	Category	Requirement
NFR-06	Traceability	Every check-in, checkout and update shall be traceable through the Visitor Log and Visitor History forms.
NFR-07	Privacy	Visitor personal data (name, mobile number) shall be visible only to profiles that need it.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Free Edition.
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Other Zoho services	E-mail delivery through the Deluge sendmail task
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

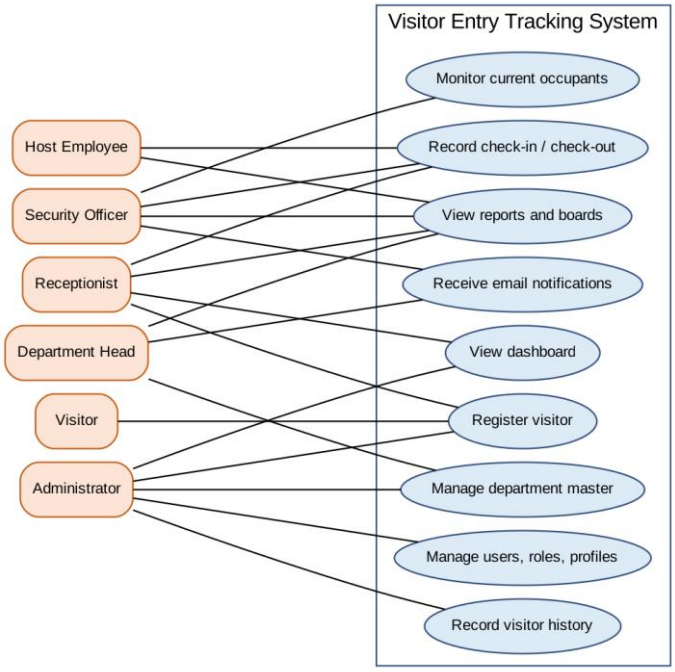


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Register Visitor

Item	Description
Use Case ID	UC-01
Use Case Name	Register Visitor
Actor(s)	Receptionist (primary); Visitor; System (generates ID and sends the department e-mail)
Pre-condition	The user is logged in with a profile that has Create permission on Visitor_Registration.
Main Flow	1. Open the Visitor Registration form; the Visitor ID is generated and locked. 2. Enter name, mobile number, purpose, person to meet, department, entry date and time and status “Inside”. 3. Submit. 4. The record is saved and the department head receives an entry e-mail.
Alternate Flow	If the department has no matching Department Master record or no contact e-mail, the record is saved and no e-mail is sent.
Post-condition	The record appears in the All Visitor Registrations report and the Visitor Registrations Board; the department head has the notification.
Related FR	FR-01, FR-02, FR-07, FR-09

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

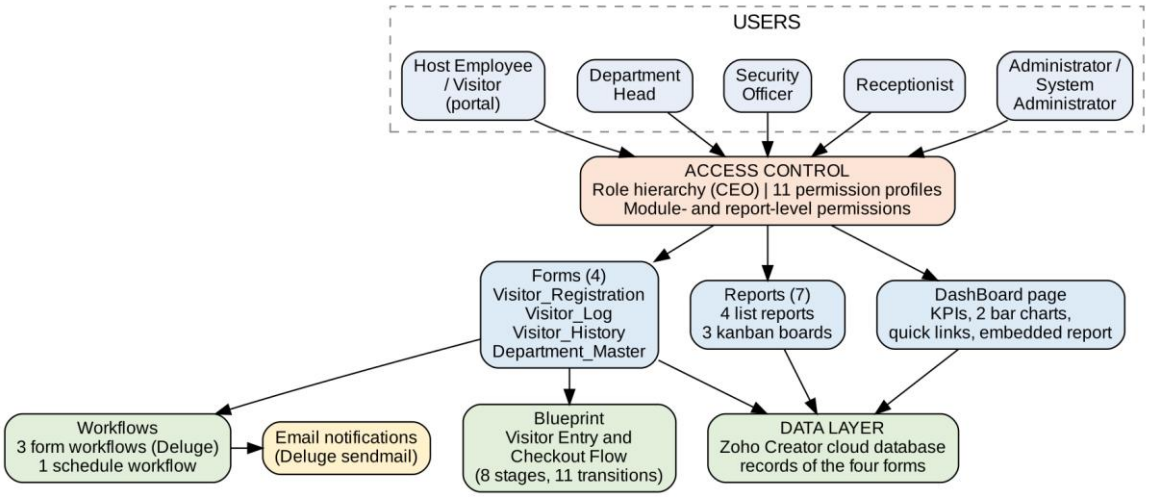


Figure 4.1: System Architecture of the Application

The application uses a layered architecture. Users sign in through a web browser (or the Zoho Creator mobile app) and are filtered by the access-control layer — the CEO role and the permission profiles. The application layer holds four forms, seven reports, the dashboard page, three form workflows, one scheduled workflow and the Blueprint; e-mails are sent through the Deluge sendmail task. All records are stored in Zoho Creator's cloud database.

4.2 Data Model

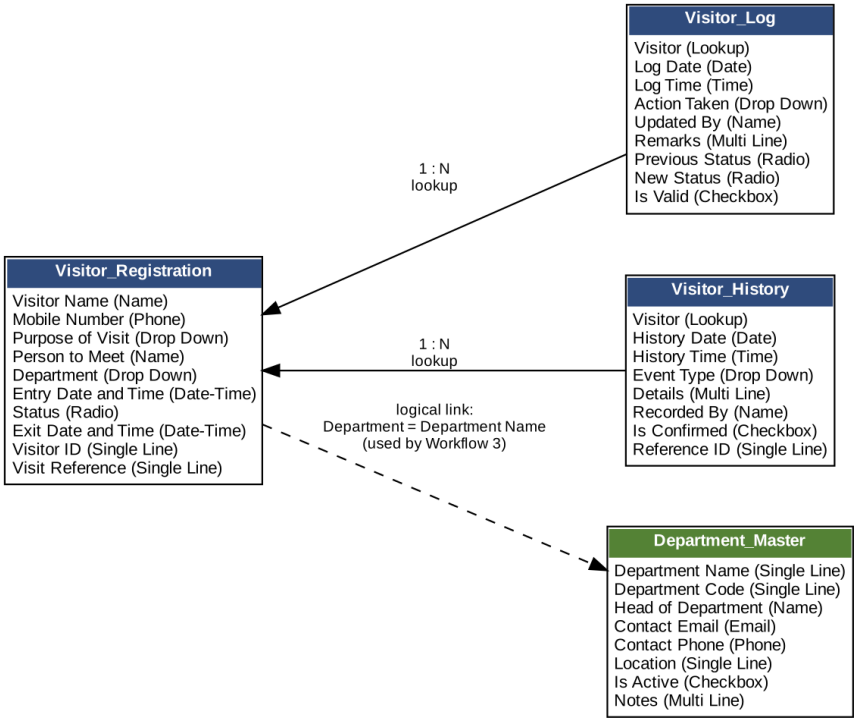


Figure 4.2: Entity–Relationship Diagram of the Application

Visitor_Registration is the central form. Visitor_Log and Visitor_History each contain a lookup field, Visitor, that points to a registration and displays its Visitor ID, giving a one-to-many relationship (one visitor, many log or history entries). Department_Master is related logically: the Department chosen in a registration is matched with the Department Name in the master by Workflow 3 to find the head of department and e-mail address. No subforms are used.

4.3 Form Design

Table 4.1: Field Specification – Visitor_Registration Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Visitor Name	visitor_name	Name	No	First name shown; prefix, last name and suffix hidden
Mobile Number	mobile_number	Phone	No	Visitor contact number
Purpose of Visit	purpose_of_visit	Drop Down	No	Meeting, Student Visit, Delivery, Official Work, Interview, Other, Personal; “others” option enabled
Person to Meet	person_to_meet	Name	No	Host’s name
Department	department	Drop Down	No	Administration, CSE, ECE, EEE, MECH, IT; “others” option enabled
Entry Date and Time	entry_datetime	Date-Time	No	Format dd-MMM-yyyy hh:mm:ss

Field Label	Link Name	Field Type	Mandatory	Remarks
Status	status	Radio	No	Inside or Checked Out; “others” option enabled
Exit Date and Time	exit_datetime	Date-Time	No	Validated against entry time (Workflow 2)
Visitor ID	visitor_id	Single Line	No	Generated by Workflow 1 on load; field disabled
Visit Reference	visit_reference	Single Line	No	Free-text reference for the visit

Table 4.2: Field Specification – Visitor_Log Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Visitor	visitor	Lookup	No	Points to Visitor_Registration; displays Visitor ID; “Add New” allowed
Log Date	log_date	Date	No	Format dd-MMM-yyyy
Log Time	log_time	Time	No	Format hh:mm:ss
Action Taken	action_taken	Drop Down	No	Check In, Checkout, Update; “others” option enabled
Updated By	updated_by	Name	No	Person who made the entry
Remarks	remarks	Multi Line	No	Notes on the action
Previous Status	previous_status	Radio	No	Inside or Checked Out
New Status	new_status	Radio	No	Inside or Checked Out
Is Valid	is_valid	Checkbox	No	Default: unchecked

Table 4.3: Field Specification – Visitor_History Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Visitor	visitor	Lookup	No	Points to Visitor_Registration; displays Visitor ID
History Date	history_date	Date	No	Format dd-MMM-yyyy
History Time	history_time	Time	No	Format hh:mm:ss
Event Type	event_type	Drop Down	No	Entry, Out, Registration; “others” option enabled
Details	details	Multi Line	No	Description of the event
Recorded By	recorded_by	Name	No	Person who recorded the event
Is Confirmed	is_confirmed	Checkbox	No	Default: unchecked
Reference ID	reference_id	Single Line	No	Reference for the event

Table 4.4: Field Specification – Department_Master Form

Field Label	Link Name	Field Type	Mandatory	Remarks
Department Name	department_name	Single Line	No	Matched with the Department chosen at registration
Department Code	department_code	Single Line	No	Short code for the department
Head of Department	head_of_department	Name	No	Addressed in the entry e-mail
Contact Email	contact_email	Email	No	Recipient of the entry notification
Contact Phone	contact_phone	Phone	No	Department phone number
Location	location	Single Line	No	Block / room
Is Active	is_active	Checkbox	No	Default: unchecked
Notes	notes	Multi Line	No	Additional information

4.4 Process and Workflow Design

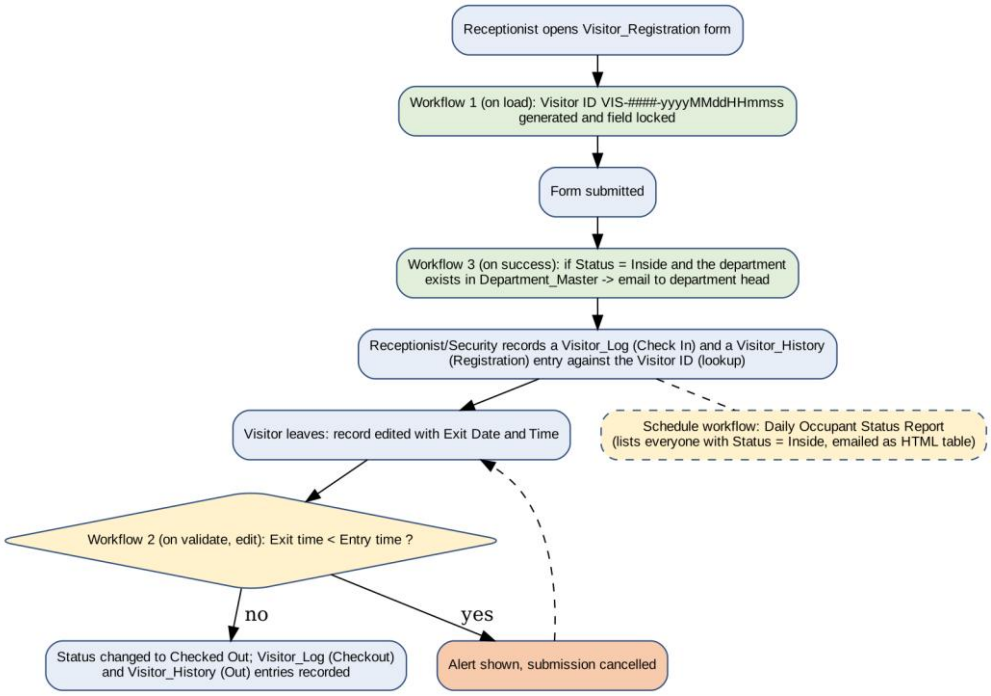


Figure 4.3: Workflow / Process Flow for a Visitor Visit

A visit starts when the registration form is opened: Workflow 1 generates and locks the Visitor ID. When the form is submitted with the status “Inside”, Workflow 3 e-mails the department head. The receptionist or security officer then records a Visitor Log (Check In) and a Visitor History entry against the Visitor ID. When the visitor leaves, the registration is edited with the exit date and time; Workflow 2 blocks the update if the exit time is earlier than the entry time. Independently, the scheduled workflow lists everyone whose status is “Inside” and e-mails the security team.

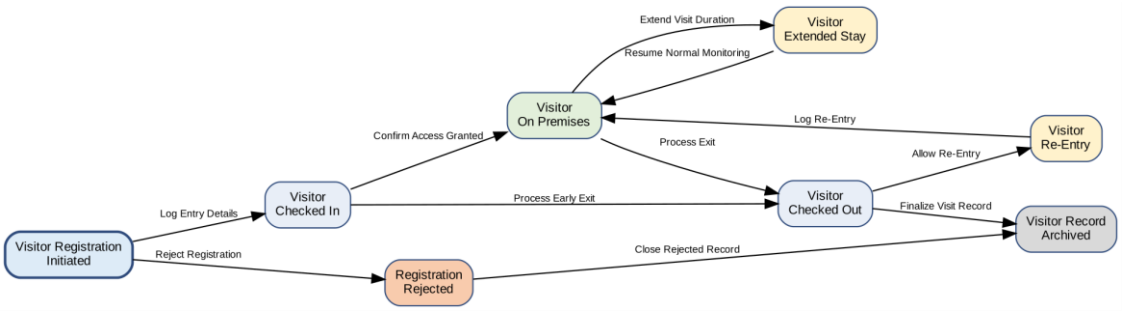


Figure 4.4: Blueprint "Visitor Entry and Checkout Flow" – Stages and Transitions

Table 4.5: Blueprint Transitions (Visitor_Registration)

Transition	From Stage	To Stage
Log Entry Details	Visitor Registration Initiated	Visitor Checked In
Reject Registration	Visitor Registration Initiated	Registration Rejected
Confirm Access Granted	Visitor Checked In	Visitor On Premises
Process Early Exit	Visitor Checked In	Visitor Checked Out
Process Exit	Visitor On Premises	Visitor Checked Out
Extend Visit Duration	Visitor On Premises	Visitor Extended Stay
Resume Normal Monitoring	Visitor Extended Stay	Visitor On Premises
Allow Re-Entry	Visitor Checked Out	Visitor Re-Entry
Log Re-Entry	Visitor Re-Entry	Visitor On Premises
Finalize Visit Record	Visitor Checked Out	Visitor Record Archived
Close Rejected Record	Registration Rejected	Visitor Record Archived

4.5 User Interface and Navigation Design

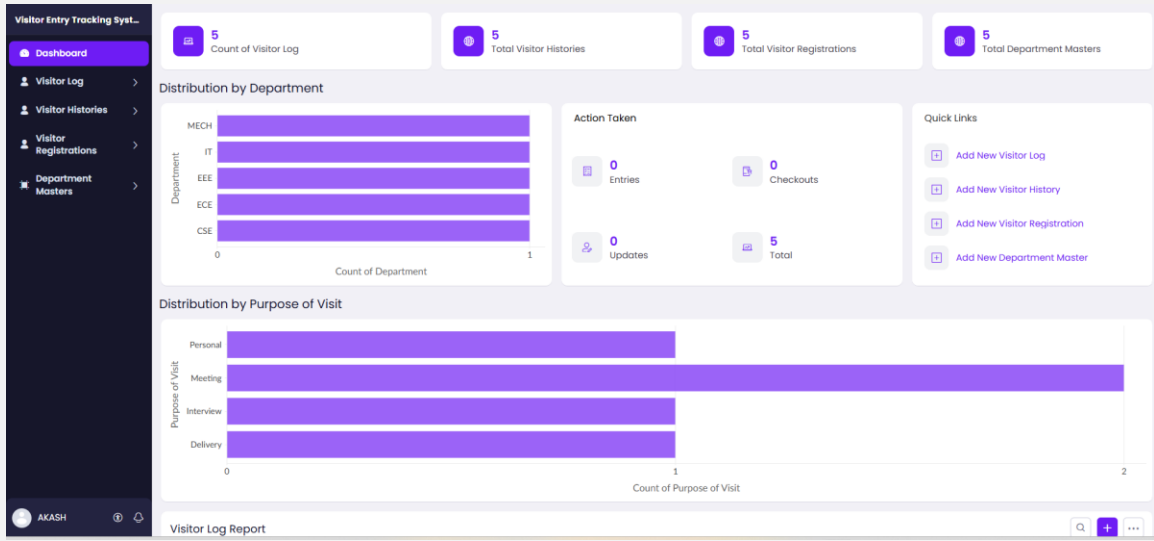


Figure 4.5: Navigation Structure of the Application and the DashBoard

The application menu lists the four forms with their reports and boards, and the DashBoard page. The dashboard shows record counts, action counts, two bar charts, the Quick Links panel with four “Add New” buttons, and the embedded Visitor Log Report.

4.6 Access Control Design

Table 4.6: Profile–Module Permission Matrix (as configured)

Profile	Visitor_ Registration	Visitor_ Log	Visitor_ History	Department_ Master
Receptionist	Create, view all, modify all, import, export	Create, view all, modify all, import, export	View all, export	View all
Security Officer	View all	Create, view all, modify all, import, export	View all, export	—
Department Head	View all	View all	View all	Create, view all, modify all, import, export
System Administrator	Full	Full	Full	Full
Host Employee (portal)	View all	Create, view all, modify all, import, export	View all	—
Visitor (portal)	Create, view all, modify all, import, export	Reports view only	Reports view only	—
Administrator	All permissions	All permissions	All permissions	All permissions

The CEO role sits at the top of the role hierarchy and can access the data of all other users. The Read and Write profiles are generic default profiles; the Developer and Customer profiles are platform defaults. Report-level permissions (View, Edit, Delete) are set per report for each custom profile. Section 7.3 notes where these permissions should be tightened.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Visitor Entry Tracking System” was created in Zoho Creator with the date format dd-MMM-yyyy and a 24-hour time format. Four forms were added, followed by their reports, the dashboard page, the workflows and the Blueprint. The left navigation menu gives access to every form, report, kanban board and the dashboard.

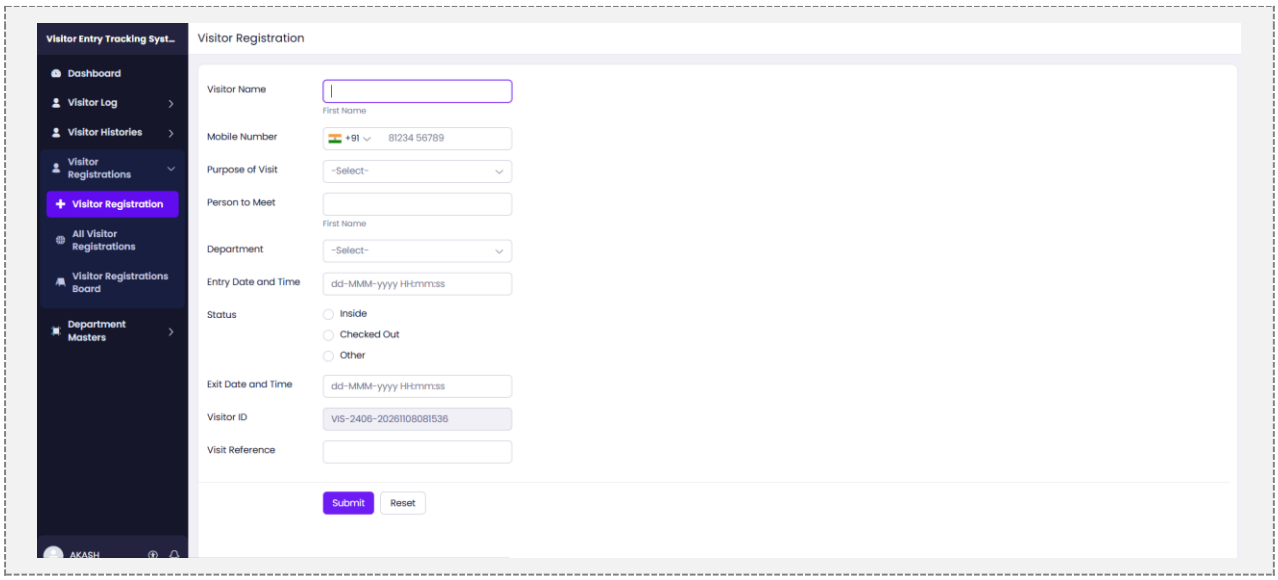


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Four forms were created: Visitor_Registration, Visitor_Log, Visitor_History and Department_Master. Together they use twelve distinct field types — Name, Phone, Email, Single Line, Multi Line, Drop Down, Radio, Checkbox, Date, Time, Date-Time and Lookup — for structured data entry. Visitor_Log and Visitor_History each contain a Lookup field to Visitor_Registration that displays the Visitor ID and allows a new registration to be added on the spot. Detailed field specifications are in Tables 4.1–4.4.

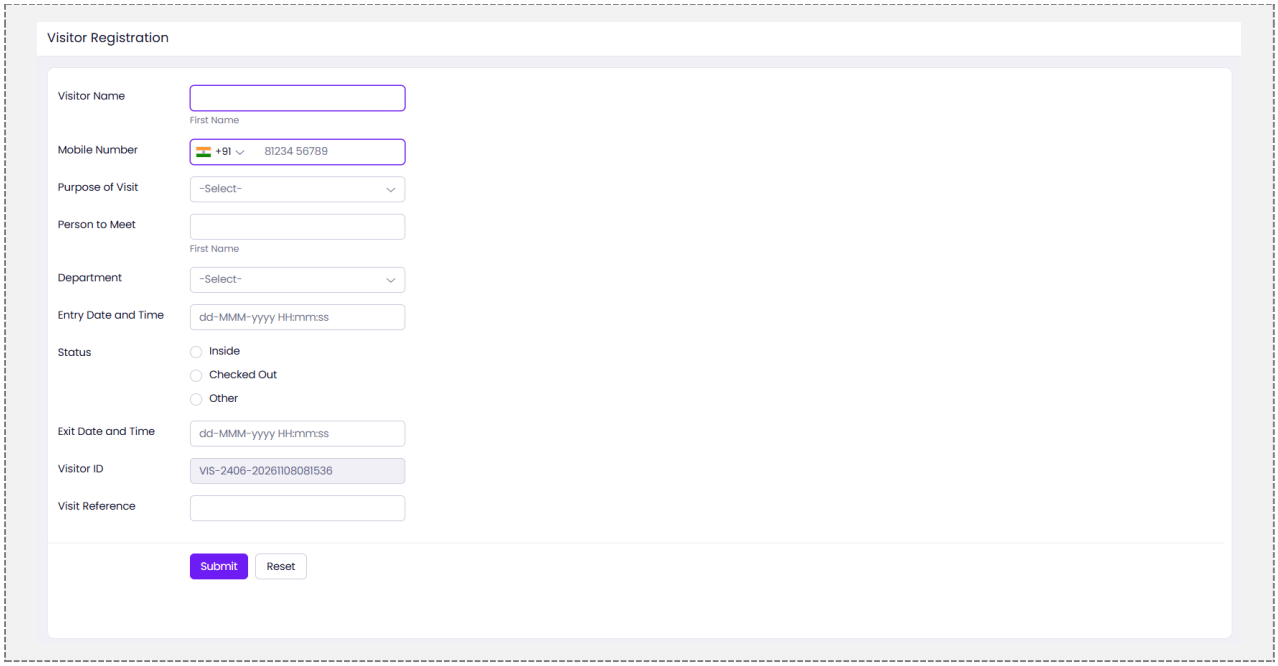


Figure 5.2: Visitor_Registration Form in the Form Builder

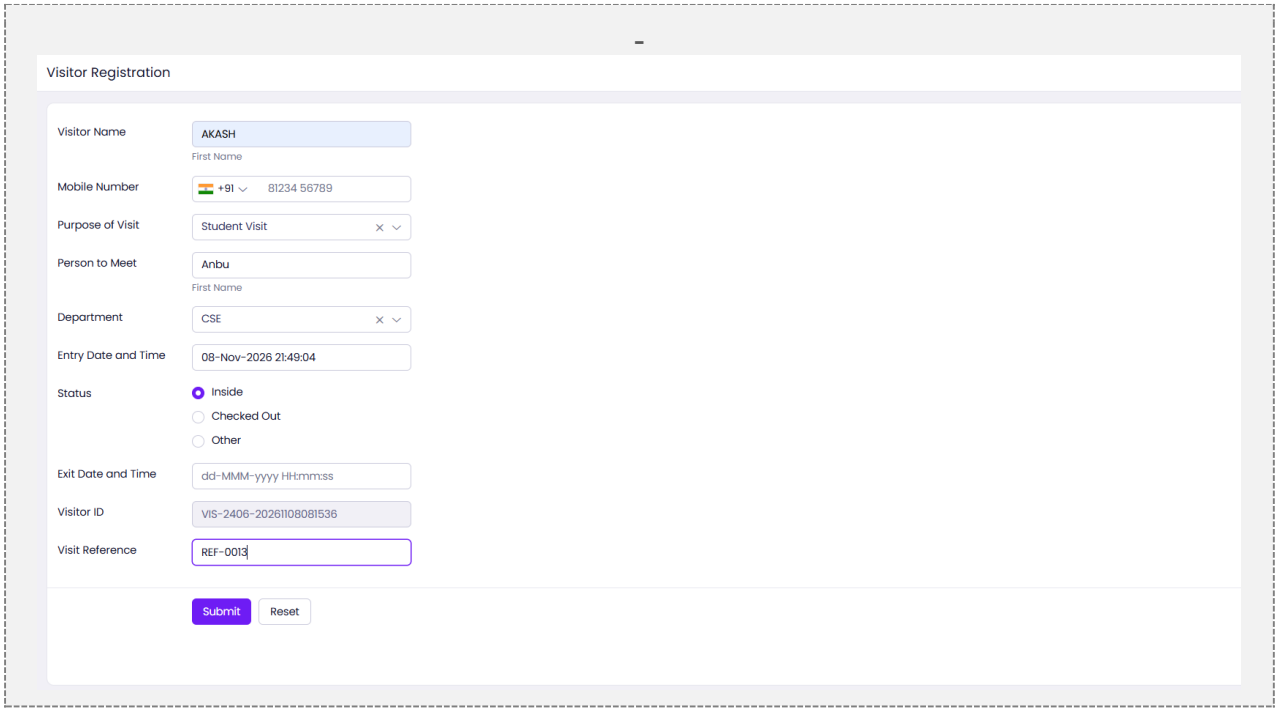


Figure 5.3: Visitor_Registration Form in Live Mode

5.3 Reports

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Formatting / Grouping	Custom Actions
All Visitor Registrations	List	Visitor_Registration	Conditional formatting on Department (3 rules)	None
Visitor Log Report	List	Visitor_Log	Conditional formatting on Action Taken (2 rules)	None
All Visitor Histories	List	Visitor_History	Conditional formatting on Event Type (2 rules)	None
All Department Masters	List	Department_Master	None	None
Visitor Registrations Board	Kanban	Visitor_Registration	Grouped by Purpose of Visit	None
Board	Kanban	Visitor_Log	Grouped by Action Taken	None
Event Types	Kanban	Visitor_History	Grouped by Event Type	None

Four list reports let users view, search and edit records according to their permissions, and three kanban boards group the records visually by purpose, action and event type. The Visitor Log Report is also embedded in the dashboard. The report quick view and detail view are configured so that opening a log entry also shows the related visitor's details.

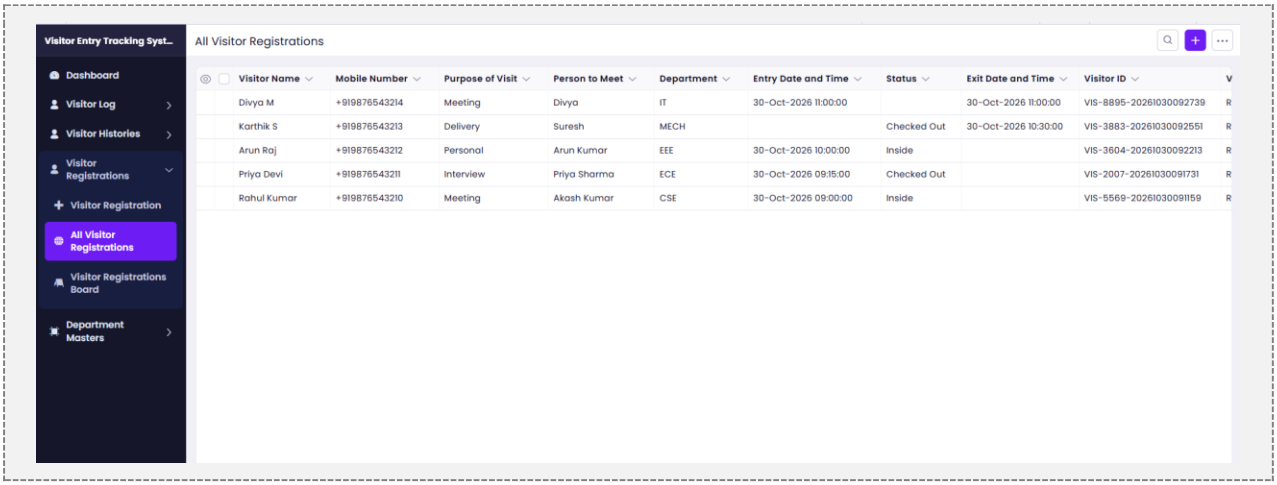


Figure 5.4: All Visitor Registrations List Report

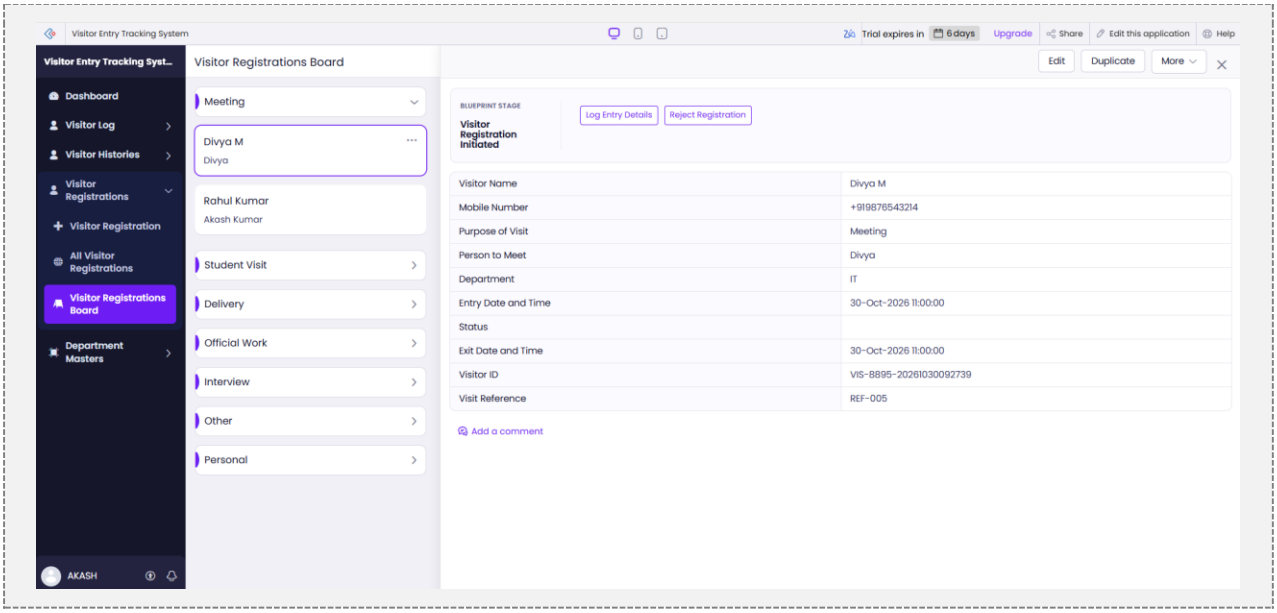


Figure 5.5: Visitor Registrations Board (Kanban Report)

5.4 Pages and Dashboards

The DashBoard page was built in the Zoho Creator page builder. It contains four record-count panels (Visitor Log, Visitor Histories, Visitor Registrations and Department Masters), an "Action Taken" panel with Entries, Checkouts, Updates and Total counts, a bar chart of visitors by department, a bar chart of visitors by purpose of visit, a Quick Links panel with four "Add New" buttons (Log, History, Registration, Department Master), and the embedded Visitor Log Report.

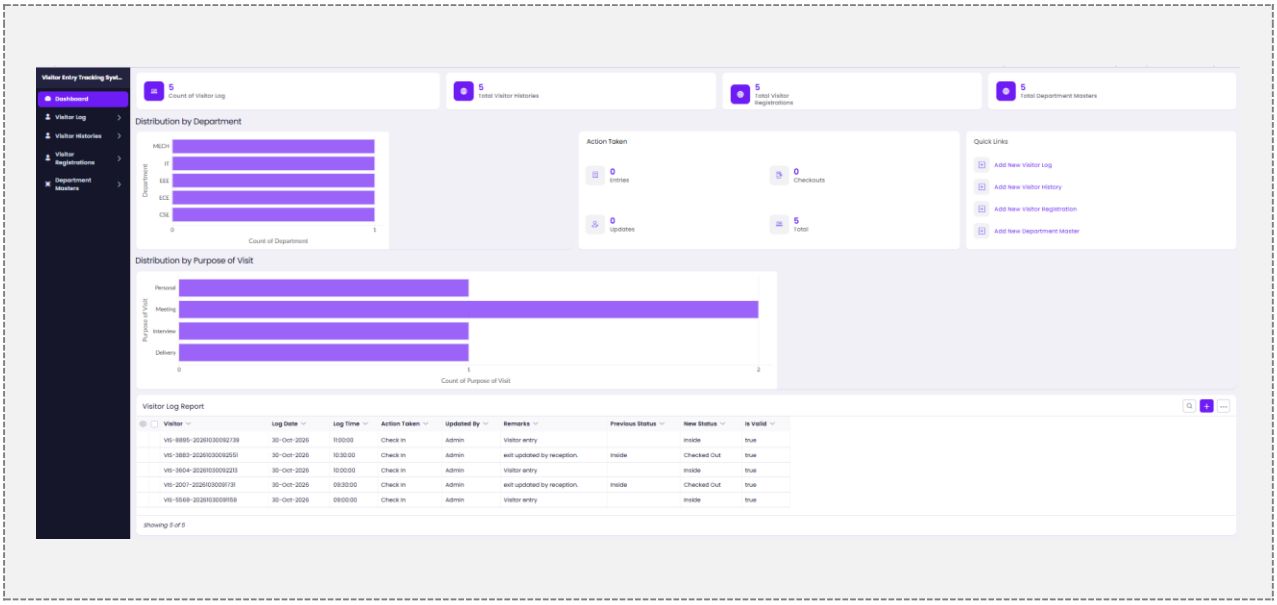


Figure 5.6: DashBoard Page

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Visitor_Registration | Workflow 1 | On Add → On Load	Generate a unique Visitor ID and lock the field	randomNumber(), now, toString() with a date pattern, string concatenation, input field assignment, disable
DS-02	Visitor_Registration | Workflow 2 | On Edit → On Validate	Reject an exit date-time earlier than the entry date-time	Nested if, null checks, date-time comparison, alert, cancel submit
DS-03	Visitor_Registration | Workflow 3 | On Add → On Success	E-mail the department head when a visitor is Inside	Record fetch by criteria, nested if, null checks, name sub-fields, string building, sendmail
DS-04	Schedule | Daily Occupant Status Report	E-mail the security team the list of visitors currently inside	Record fetch, count(), if/else, for each loop, concat(), HTML e-mail, zoho.currentdate

The full source code of all four scripts is in Appendix C. The two scripts that are central to data integrity are shown below.

Code 5.1: Visitor ID generation (DS-01, Workflow 1)

```
// DS-01 | Form: Visitor_Registration | Event: On Add → On Load
// Generate unique visitor ID using current timestamp and random number
randomValue = randomNumber(1000,9999);
currentDateTime = now;
visitorID = "VIS-" + randomValue.toString() + "-" +
currentDateTime.toString("yyyyMMddHHmmss");
// Set the generated visitor ID to the visitor_id field
input.visitor_id = visitorID;
// Disable the visitor_id field so users cannot edit it
disable visitor_id;
```

Code 5.2: Exit-time validation (DS-02, Workflow 2)

```
// DS-02 | Form: Visitor_Registration | Event: On Edit → On Validate
// Validate that exit time is not before entry time
if(input.exit_datetime != null && input.entry_datetime != null)
{
    if(input.exit_datetime < input.entry_datetime)
    {
        alert "Error: Exit time cannot be before entry time. Please correct the times
before submitting.";
        cancel submit;
    }
}
```

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Action(s)	Script Ref.
Auto-Generate Visitor ID	Form Workflow (Deluge)	Visitor_Registration → on add → on load	Builds "VIS-####-yyyyMMddHHmmss", fills Visitor ID and disables the field	DS-01
Validate Exit Time Logic	Form Workflow (Deluge)	Visitor_Registration → on edit → on validate	If exit < entry, shows an alert and cancels the submission	DS-02
Notify Department Head on Entry	Form Workflow (Deluge, Send Mail)	Visitor_Registration → on add → on success	If Status = Inside and the department has an e-mail, e-mails the head of department	DS-03
Daily Occupant Status Report	Schedule	Date-time schedule; time zone Asia/Kolkata	Collects all visitors with Status = Inside and e-mails an HTML summary to the security team	DS-04
Visitor Entry and Checkout Flow	Blueprint	Visitor_Registration records	Moves a record through 8 stages using 11 transitions	—

Workflows 1 to 3 are form workflows; Workflow 1 runs when the add form loads, Workflow 3 only after a successful submission, and Workflow 2 when an edit is validated. The Blueprint stages (Table 4.5) are configured independently of the Status field. No approval process or batch workflow is used.

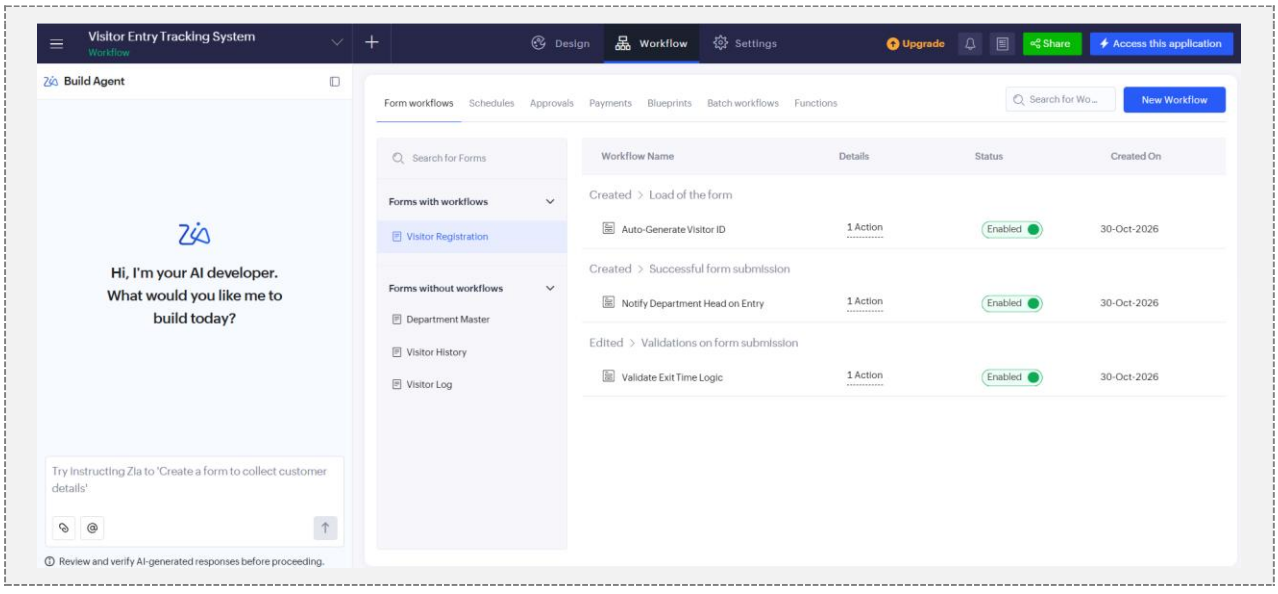


Figure 5.7: Workflow Configuration

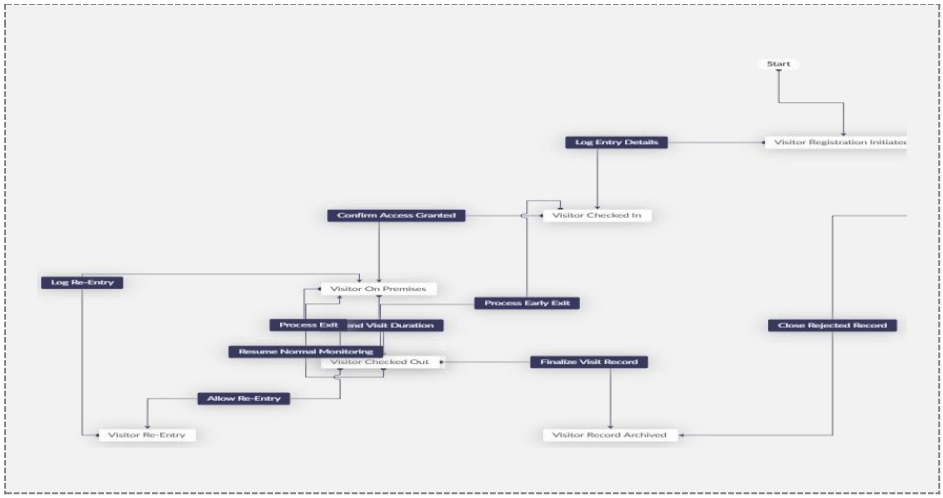


Figure 5.8: Blueprint Configuration

5.7 User Management and Role-Based Access

Access is managed through a CEO role in the role hierarchy and eleven permission profiles. Six custom profiles — Receptionist, Security Officer, Department Head, System Administrator, Host Employee and Visitor — carry module-level permissions (Create, View all, Modify all, Import, Export) and report-level permissions (View, Edit, Delete); the remaining profiles are the Zoho Creator defaults. The matrix is shown in Table 4.6.

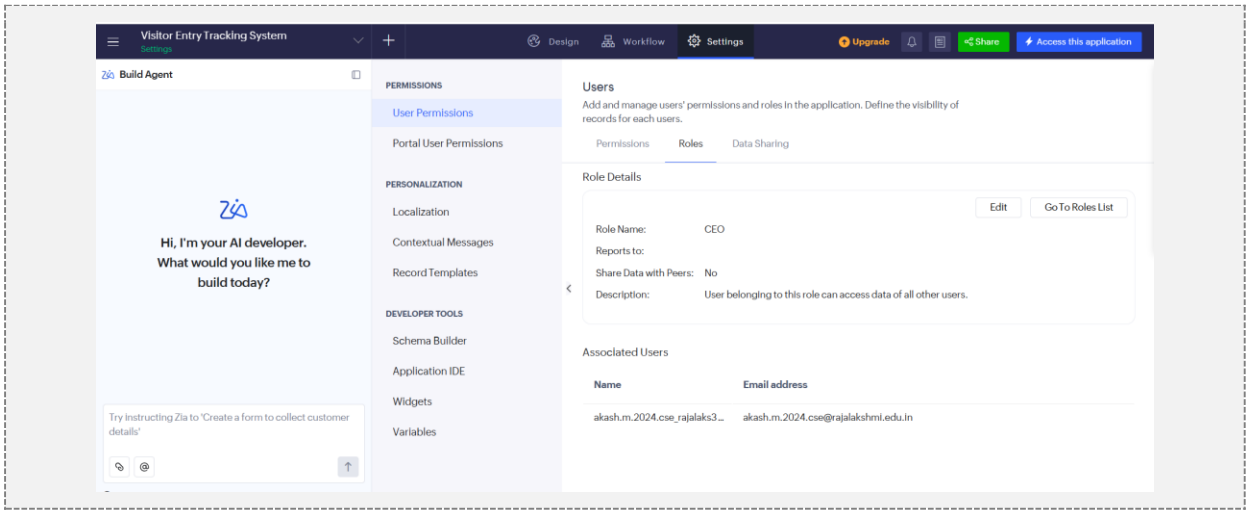


Figure 5.9: Roles and Permission Profiles Configuration

5.8 Analytics and AI Features

Analytics: The DashBoard presents two bar charts — Distribution by Department and Distribution by Purpose of Visit — both built on the Visitor_Registration form with criteria that exclude empty values, together with record-count and action-count panels (Section 5.4).

AI: No Zia or AI field was used in this version of the application. Possible uses are discussed in Section 9.2.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing follows a bottom-up approach. First the individual forms and fields are tested, then each of the three form workflows and the scheduled workflow with valid and invalid input, then the Blueprint transitions, reports and dashboard, and finally role-based access using test users for each profile rather than only the owner account. Each test case is mapped to a functional requirement in Table 3.2, and the actual result and status are recorded after execution.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC01	FR-02	Visitor ID generation	Open the Visitor Registration form.	Visitor ID appears in the format VIS-####-yyyyMMddHHmmss and the field is disabled.	Fill	Pass
TC02	FR-01, FR-07	Registration and department e-mail	Register a visitor with Status = Inside and Department = CSE (CSE exists in Department Master with an e-mail).	Record saved; department head receives the entry e-mail.	Fill	Pass
TC03	FR-03	Invalid exit time	Edit a record and set exit date-time earlier than entry date-time.	Alert shown; update cancelled.	Fill	Pass
TC04	FR-03	Valid exit time	Edit a record, set a later exit time and Status = Checked Out.	Record updated successfully.	Fill	Pass
TC05	FR-04, FR-14	Visitor Log entry via lookup	Add a log entry, choose a Visitor ID, Action Taken = Check In.	Entry saved and shown in Visitor Log Report with the visitor's details in the detail view.	Fill	Pass
TC06	FR-05	Visitor History entry	Add a history entry, Event Type = Registration, tick Is Confirmed.	Entry saved and shown in All Visitor Histories.	Fill	Pass
TC07	FR-06	Department Master	Add a department with head, e-mail, phone and Is Active ticked.	Record saved and shown in All Department Masters.	Fill	Pass
TC08	FR-11	Blueprint flow	Move a record: Initiated → Checked In → On Premises → Checked Out → Archived.	Each transition succeeds and the stage updates.	Fill	Pass
TC09	FR-08	Daily occupant summary	Activate the schedule, run it with at least one visitor Inside.	Security mailbox receives an HTML table of occupants.	Fill	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC10	FR-09, FR-10	Reports and boards	Open all list reports and kanban boards.	Records display; boards group by the chosen field; formatting applied.	Fill	Pass
TC11	FR-12	Dashboard	Open the DashBoard after entering sample data.	Counts, both charts, Quick Links and embedded report display correctly.	Fill	Pass
TC12	FR-13	Role permissions	Log in as Receptionist, Security Officer and Department Head test users.	Each user can use only the modules and reports permitted in Table 4.6.	Fill	Pass

6.3 Debugging Deluge Scripts

The four Deluge scripts were written in the Zoho Creator script editor, which checks syntax when a script is saved. Each script was then exercised with valid and invalid input from the live form. During debugging, an info statement can be placed after a fetch or an assignment to show the value in the execution log, and the workflow's execution history shows whether a script ran and where it failed. The typical problems met while developing this application were (a) matching the exact dropdown value in a criteria expression and (b) reading the sub-fields of a Name field (first_name, last_name) instead of the whole field.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Proposed / Applied Fix	Status
BUG-01	DashBoard "Entries" count shows 0.	Panel criteria use action_taken == "Entry", but the Action Taken dropdown contains "Check In".	Change the criteria to "Check In" (or add "Entry" to the dropdown).	Open
BUG-02	Some conditional-formatting rules never apply.	Rules test "Entry" (Visitor Log), "Modification" (Visitor History) and "Computer Science" (Registrations), which are not dropdown values ("Check In", "Out", "CSE").	Edit the rules to use the actual dropdown values.	Open
BUG-03	Incomplete records can be saved.	No field is marked mandatory in any form.	Make Visitor Name, Mobile Number, Purpose, Department and Entry Date-Time mandatory.	Open
BUG-04	Entry time and generated ID may be offset from local time.	Application time zone is America/Los_Angeles while the schedule uses Asia/Kolkata.	Set the application time zone to Asia/Kolkata.	Open
BUG-05	Daily summary is not delivered.	The schedule is inactive in the exported definition, starts on a past date and e-mails a placeholder address (security@company.com).	Set the real security mailbox, set a start time and activate the schedule.	Open

Bug ID	Description	Root Cause	Proposed / Applied Fix	Status
BUG-06	No e-mail is sent for some departments.	Workflow 3 matches the registration Department value with Department Name in the master; a mismatch or missing e-mail skips the e-mail.	Use identical names in the master or convert Department to a lookup.	Open
BUG-07	Status, log entries and Blueprint stage can disagree.	Status is a separate radio field; log/history entries are made manually; the Blueprint is not tied to Status.	Add a workflow to create log/history entries and synchronise Status with the Blueprint stage.	Open

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User Role	Tested	Feedback	Action Taken
Receptionist	Priya S, 30-Oct-2026	Registration form is easy to fill and the Visitor ID is generated automatically. Entering Visitor Log entries separately is extra work.	Noted as future enhancement: create log entries automatically by workflow.
Security Officer	Karthik R, 30-Oct-2026	Registration form is easy to fill and the Visitor ID is generated automatically. Entering Visitor Log entries separately is extra work.	Status field and daily occupant email added; filter by “Inside” explained in the user manual.
Department Head	Divya M, 30-Oct-2026	The email telling me a visitor has arrived is useful. I did not get it when my department name was spelled differently.	Department names in Department Master matched exactly with the dropdown values.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The Visitor Entry Tracking System was developed and tested in the Zoho Creator development environment. Forms, reports, workflows, the Blueprint, permissions and the dashboard were configured and tested before the application was published through Zoho Creator and made available to authorised users according to their roles and permissions.

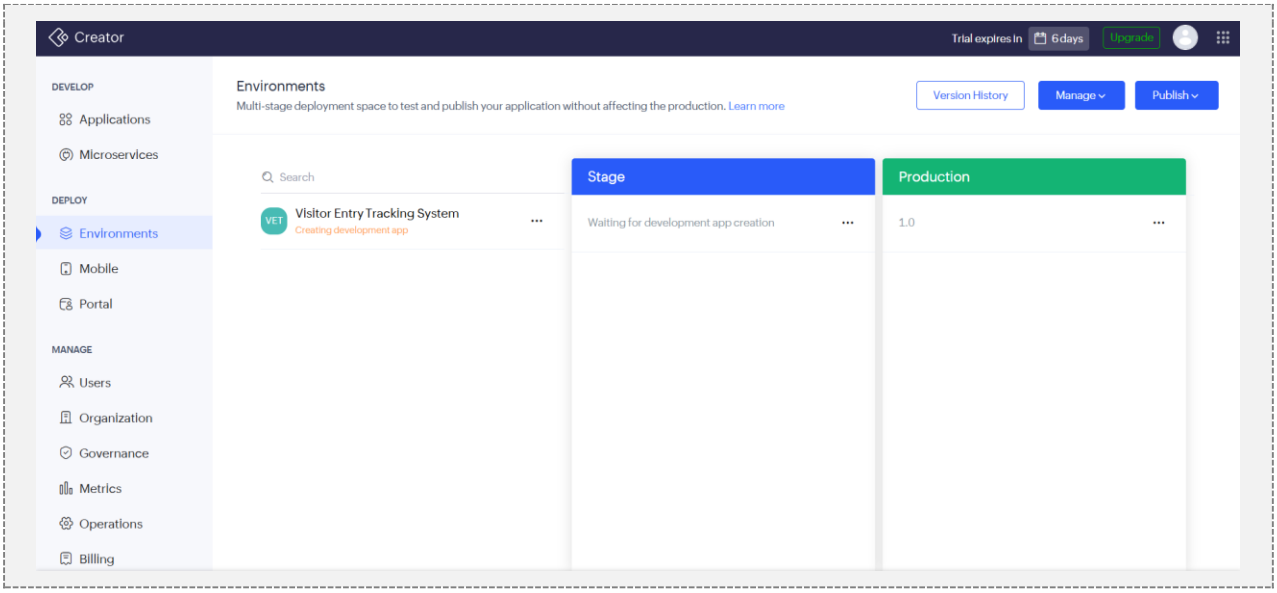


Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Real recipient e-mail addresses set for notifications and the schedule	✓
5	Application time zone verified (Asia/Kolkata)	✓
6	Schedule and workflows enabled	✓
7	Application published and URL verified	✓
8	Version / release notes recorded	✓

7.2 Performance

No formal performance measurement was carried out for this version. The design choices that keep the application responsive are: the registration form stores everything in a single form so one record per visit is enough; the Visitor Log and Visitor History forms reference the registration through a lookup rather than copying its data; the dashboard charts filter out empty values before counting; list reports show only the fields needed by the front-desk user; and Workflow 3 fetches a single registration by its ID and a single department by name instead of scanning all records. The daily summary query is limited to records whose status is “Inside”.

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role hierarchy (CEO) and eleven permission profiles	Settings → User Permissions	Unauthorised viewing or modification of visitor records
Module-level and report-level permissions for custom profiles	Receptionist, Security Officer, Department Head, System Administrator, Host Employee, Visitor	Users performing actions outside their duties (for example, a Department Head editing logs)
Invitation-based user access	Manage → Users → Add Users	Access by unauthorised persons
Locked Visitor ID field	Workflow 1 (disable visitor_id)	Tampering with or duplicating the visitor identifier
Exit-time validation	Workflow 2	Inconsistent or falsified entry/exit times

Points to tighten: the Receptionist, Security Officer and System Administrator profiles allow “Delete” on some reports, and the Visitor portal profile allows viewing and modifying all registrations. For production use, deletion should be limited to administrators and the Visitor profile should be restricted to the visitor’s own records through record-level criteria.

7.4 Governance and Administration

The application uses role-based access control so that every role has a defined scope: the CEO sees all data, the Receptionist and Security Officer operate the gate, the Department Head maintains department contacts, and the System Administrator configures the system. The Visitor Log and Visitor History forms give a business-level trail of every check-in, checkout and update with the person who recorded it. Reports and the dashboard let administrators monitor visitor activity. Platform-level audit trail settings were not configured in this version.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The scenario follows one visitor — for example, a guest visiting the CSE department. Replace this sentence with the actual sample record you used: name, purpose, department and entry time. The receptionist opens the Visitor Registration form, where the Visitor ID is generated and locked, enters the details with the status “Inside” and submits. The department head receives an entry e-mail. A Visitor Log entry (Check In) and a Visitor History entry (Registration) are added against the Visitor ID. The record then appears on the Visitor Registrations Board and in the dashboard charts and counts. When the visitor leaves, the registration is edited with the exit time and the status “Checked Out”; an exit time earlier than the entry time is rejected.

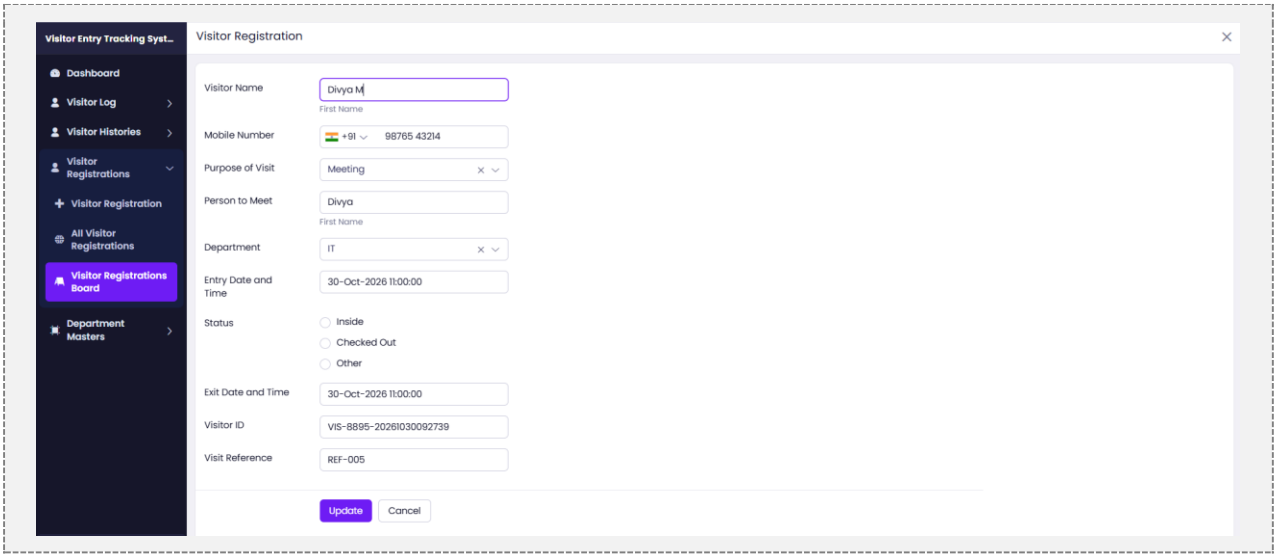


Figure 8.1: Scenario Step 1 – Registering a Visitor (auto-generated Visitor ID)

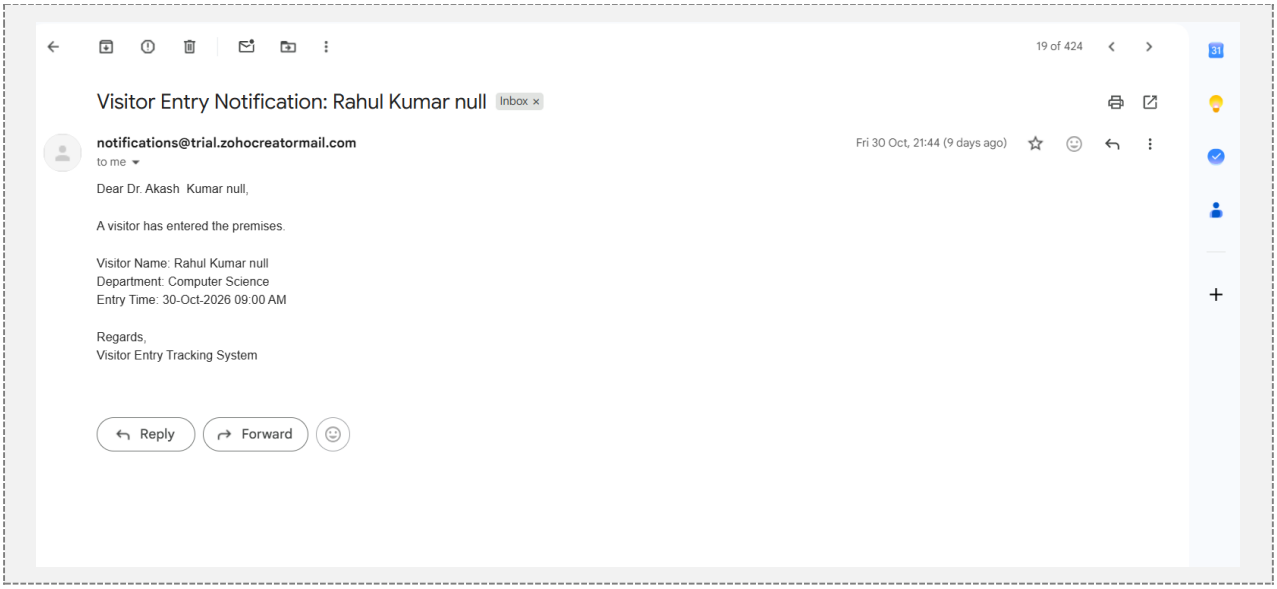


Figure 8.2: Scenario Step 2 – Department Head Notification E-mail

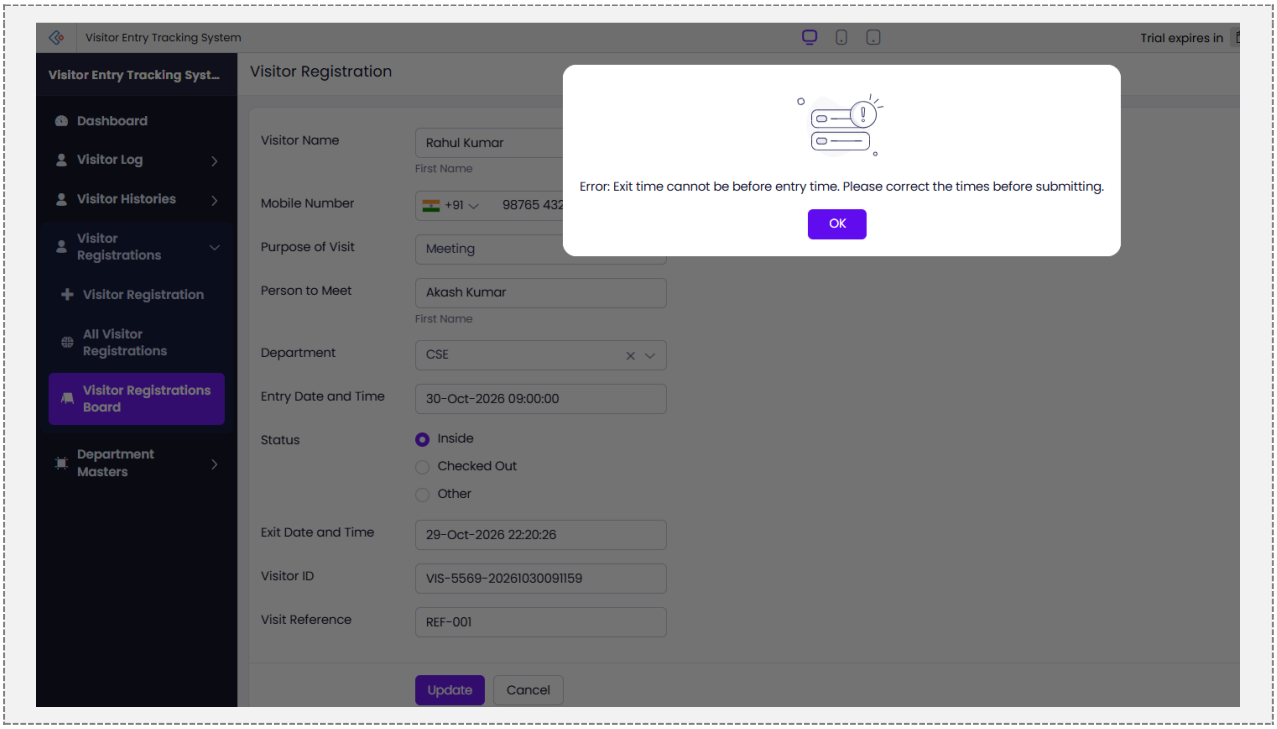


Figure 8.3: Scenario Step 3 – Exit-Time Validation Alert

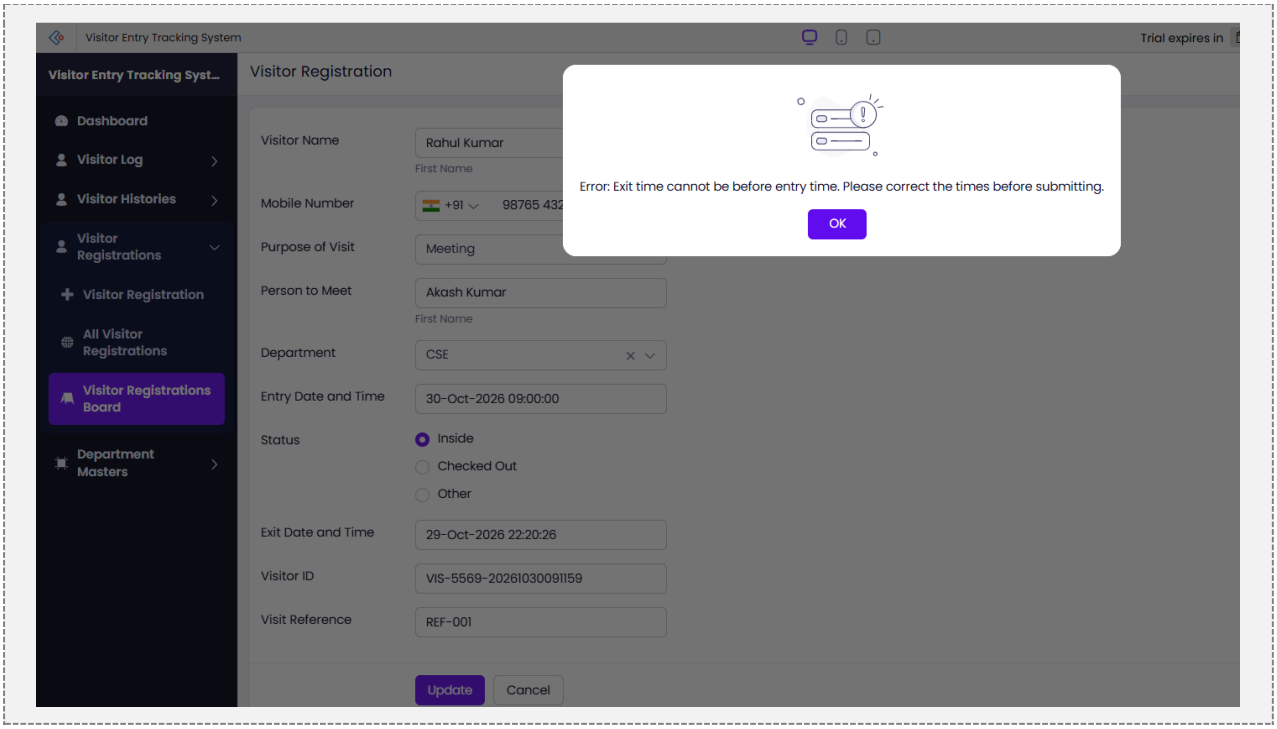


Figure 8.4: Scenario Step 4 – Daily Occupant Summary E-mail

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Objective 1: design the forms	Sections 4.3 and 5.2	Four forms created using twelve field types, with two lookup relationships (Tables 4.1 to 4.4)
Objective 2: automate with workflows	Sections 5.5 and 5.6	Three form workflows, one scheduled workflow and one Blueprint configured (Tables 5.2, 5.3, 4.5)
Objective 3: reports, dashboard and role-based access	Sections 5.3, 5.4 and 5.7	Seven reports, one dashboard, a CEO role and eleven permission profiles configured
Objective 4: e-mail notifications	Sections 5.5 and 5.6	Department-head entry e-mail and daily occupant summary configured through sendmail

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording visitors	Handwritten paper register	Registration form with typed fields and a unique Visitor ID	Legible, searchable, uniquely identified records
Notifying hosts	Phone call by security, if at all	Entry e-mail to the department head sent by a workflow	Less manual communication

Parameter	Existing System	Proposed Application	Improvement
Exit time	Often missing; never checked	Recorded on the registration; validated against entry time	Fewer incorrect or missing times
Who is inside?	Counted by hand	Status field, kanban boards and a daily e-mailed occupant table	Faster, safer awareness of occupancy
Summaries	Counted by hand	Dashboard counts and charts by department and purpose	Instant visual summary
Access control	Anyone with the register can view or change it	CEO role and eleven permission profiles	Controlled, role-based access

8.4 Limitations of the Low-Code Approach

Zoho Creator limits what can be customised: layouts, validations and automation have to be expressed through the available form, workflow and Deluge capabilities, and some requirements — such as printing visitor badges or reading an ID card — would need extensions. Platform features such as time-zone, permission and schedule settings must also be configured carefully, as the defect log (Table 6.2) shows. In this application, status fields, log entries and the Blueprint stage are not automatically synchronised, and no field is mandatory, so data quality depends on how the forms are used until these gaps are closed.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application was built with no-code configuration: forms, lookups, list and kanban reports, the dashboard page, the Blueprint and the permission profiles. Low-code Deluge was used in four scripts for logic the visual tools do not provide — generating the ID, comparing date-times, fetching related records and composing e-mails. No pro-code extension (widgets, Catalyst functions or external APIs) was needed, which shows that a complete, useful application can be delivered by a citizen developer; pro-code would become relevant for badge printing, hardware integration or high-volume analytics.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Visitor Entry Tracking System was developed on Zoho Creator to register visitors, log their entry and exit, keep a history of events and maintain department contacts in one place. Four forms collect the data; seven reports and a dashboard display it; three form workflows generate the Visitor ID, validate the exit time and notify the department head; a scheduled workflow e-mails the security team a daily list of occupants; a Blueprint guides each visit through eight stages; and a CEO role with eleven permission profiles controls access. The project shows how a low-code platform can replace a paper register with an organised, traceable system, while the defect log identifies the refinements needed before large-scale use.

9.2 Future Enhancements

- Make key fields mandatory, and convert the Department drop-down into a lookup to Department_Master so that departments always match.
- Create Visitor_Log and Visitor_History entries automatically by workflow when a registration is added or its status changes, and synchronise the Blueprint stage with the Status field.
- Correct the dashboard and conditional-formatting criteria, set the application time zone to Asia/Kolkata, and activate the daily schedule with the real security mailbox.
- Restrict the Visitor portal profile to a visitor's own records and disable deletion for operational profiles.
- Add visitor photo and ID-proof upload, QR-coded visitor passes, SMS or WhatsApp alerts to the host, pre-registration through the portal with host approval, and a blacklist check.
- Add AI features, for example Zia-based text analysis of remarks, and analytics on peak visiting hours.

9.3 Learning Outcomes and Career Relevance

This project gave practical experience in building a complete application by configuring forms, relationships, workflows and permissions instead of writing extensive code. It showed the importance of data modelling (lookups), validation, role-based access and testing with real user roles — and the value of reviewing the finished application critically for gaps.

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[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/ (Accessed).

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Visitor Entry Tracking System application and navigation menu	5.1	Figure 5.1
II	Forms with ≥ 8 distinct field types	Y	All four forms (12 field types)	5.2	Tables 4.1–4.4
II	Lookup field (relationship between forms)	Y	Visitor_Log.Visitor and Visitor_History.Visitor → Visitor_Registration	4.2, 5.2	Figure 4.2
II	Subform (parent–child data)	N	—	4.2, 5.2	—
II	Field validations / layout rules	Y	Workflow 2 – exit time validation (DS-02); locked Visitor ID (DS-01)	5.5	Code 5.1, 5.2
II	List report with filter, grouping, custom action	Partial	Four list reports with conditional formatting; no filter, grouping or custom action	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	Y	Three kanban boards	5.3	Figure 5.5
II	Pivot table / chart report	Partial	Two bar charts on the dashboard (not stand-alone chart reports)	5.4	Figure 5.6
II	Page / dashboard built with UI Builder	Y	DashBoard with counts, two charts, quick links, embedded report	5.4	Figure 5.6
II	Deluge – form events (validate / success / field actions)	Y	On load (DS-01), on validate (DS-02), on success (DS-03)	5.5	Table 5.2
II	Deluge – functions, fetch, loops, collections	Partial	Record fetch and for each loop used; no custom functions or collections	5.5	Appendix C
II	Email / in-app notifications	Y	Workflow 3 and the daily schedule (sendmail)	5.5, 5.6	Table 5.3
II	Form workflow	Y	Three form workflows	5.6	Table 5.3
II	Scheduled workflow	Y	Daily Occupant Status Report (activate before demo)	5.6	Table 5.3
II	Approval process	N	Not used	5.6	—
II	Blueprint	Y	Visitor Entry and Checkout Flow (8 stages, 11 transitions)	5.6	Table 4.5
II	Batch / report workflow	N	Not used (Batch: 0)	5.6	—
II	Testing & debugging evidence	Partial	Test cases and defect log prepared; results to be recorded	6	Tables 6.1, 6.2
III	Integration with a Zoho service	N	Not used	5.7	—
III	Connection / REST API call (invokeurl)	N	Not used	5.7	—
III	Users, roles and permission profiles	Y	CEO role; eleven permission profiles	5.7	Table 4.6

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Record-level access criteria / customer portal	Partial	Portal profiles (Host Employee, Visitor) defined; no record-level criteria	5.7	Table 4.6
III	Zia / AI feature	N	Not used	5.8	—
III	Analytics (charts / Zoho Analytics)	Y	Dashboard bar charts	5.8	Figure 5.6
IV	Performance optimisation applied & measured	N	Design choices only; not measured	7.2	—
IV	Deployment across environments	Y	Development-Protuction	7.1	Figure 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Roles and profiles	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Audit trail not configured	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Not used	8.5	—
V	Low-code limitations & spectrum analysis	Y	Sections 8.4, 8.5	8.4, 8.5	—

Total features fully implemented: 15 / 30 (partially implemented: 5; deployment row to be confirmed).

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7, 5.8		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 8.4, 8.5, 9.2, 9.3		

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Auto-Generate Visitor ID (DS-01)

Location: Visitor_Registration → Workflow "Auto-Generate Visitor ID" → On Add → On Load

```
// Generate unique visitor ID using current timestamp and random number
randomValue = randomNumber(1000,9999);
currentDateTime = now;
visitorID = "VIS-" + randomValue.toString() + "-" +
currentDateTime.toString("yyyyMMddHHmmss");
// Set the generated visitor ID to the visitor_id field
input.visitor_id = visitorID;
// Disable the visitor_id field so users cannot edit it
disable visitor_id;
```

C.2 Validate Exit Time Logic (DS-02)

Location: Visitor_Registration → Workflow "Validate Exit Time Logic" → On Edit → On Validate

```
// Validate that exit time is not before entry time
if(input.exit datetime != null && input.entry datetime != null)
{
    if(input.exit_datetime < input.entry_datetime)
    {
        alert "Error: Exit time cannot be before entry time. Please correct the times
before submitting.";
        cancel submit;
    }
}
```

C.3 Notify Department Head on Entry (DS-03)

Location: Visitor_Registration → Workflow "Notify Department Head on Entry" → On Add → On Success

```
// Fetch visitor registration record
visitorRecord = visitor_registration[ID == input.ID];
// Check visitor status
if(visitorRecord.status == "Inside")
{
    // Get department name
    departmentName = visitorRecord.department;
    // Check department is not empty
    if(departmentName != null && departmentName != "")
    {
        // Fetch department master record
        departmentRecord = department_master[department_name == departmentName];
        // Check whether department record exists
        if(departmentRecord != null)
        {
            // Check department email exists
            if(departmentRecord.contact_email != null && departmentRecord.contact_email !=
"")
            {
                // Get visitor name
                visitorFirstName = visitorRecord.visitor_name.first_name;
                visitorLastName = visitorRecord.visitor_name.last_name;
                visitorFullName = visitorFirstName + " " + visitorLastName;
                // Get department head name
                headFirstName = departmentRecord.head_of_department.first_name;
                headLastName = departmentRecord.head_of_department.last_name;
                headFullName = headFirstName + " " + headLastName;
                // Prepare email
                emailSubject = "Visitor Entry Notification: " + visitorFullName;
                emailMessage = "Dear " + headFullName + ",\n\n";
                emailMessage = emailMessage + "A visitor has entered the premises.\n\n";
                emailMessage = emailMessage + "Visitor Name: " + visitorFullName + "\n";
```

```
                emailMessage = emailMessage + "Department: " + departmentName + "\n";
                emailMessage = emailMessage + "Entry Time: " +
visitorRecord.entry_datetime.toString("dd-MMM-yyyy hh:mm a") + "\n\n";
                emailMessage = emailMessage + "Regards,\nVisitor Entry Tracking System";
                // Send notification
                sendmail
                [
                    from :zoho.loginuserid
                    to :departmentRecord.contact_email
                    subject :emailSubject
                    message :emailMessage
                    content type :Plain Text
                ]
            }
        }
    }
}
```

C.4 Daily Occupant Status Report (DS-04)

Location: Schedule → “Daily Occupant Status Report” → On Start

```
// Fetch all visitors currently inside the building
occupants = visitor_registration[status == "Inside"];
// Check if there are any occupants
if(occupants.count() > 0)
{
    // Build the summary message
    summaryMessage = "<h2>Daily Occupant Summary</h2>";
    summaryMessage = summaryMessage.concat("<p><strong>Date:</strong> " +
zoho.currentdate.toString() + "</p>");
    summaryMessage = summaryMessage.concat("<p><strong>Total Occupants:</strong> " +
occupants.count().toString() + "</p>");
    summaryMessage = summaryMessage.concat("<h3>Current Occupants List:</h3>");
    summaryMessage = summaryMessage.concat("<table border='1' cellpadding='5'
cellspacing='0'>");
    summaryMessage =
summaryMessage.concat("<tr><th>Name</th><th>Department</th><th>Mobile</th><th>Purpose</th><
/tr>");
    // Iterate through each occupant and add to the message
    for each  occupant in occupants
    {
        name = occupant.visitor_name.first_name.concat("
").concat(occupant.visitor_name.last_name);
        department = occupant.department;
        mobile = occupant.mobile_number;
        purpose = occupant.purpose_of_visit;
        summaryMessage = summaryMessage.concat("<tr>");
        summaryMessage = summaryMessage.concat("<td>" + name + "</td>");
        summaryMessage = summaryMessage.concat("<td>" + department + "</td>");
        summaryMessage = summaryMessage.concat("<td>" + mobile + "</td>");
        summaryMessage = summaryMessage.concat("<td>" + purpose + "</td>");
        summaryMessage = summaryMessage.concat("</tr>");
    }
    summaryMessage = summaryMessage.concat("</table>");
    // Send email to security team
    sendmail
    [
        from :zoho.loginuserid
        to :"security@company.com"
        subject :"Daily Occupant Summary - " + zoho.currentdate.toString()
        message :summaryMessage
        content type :HTML
    ]
}
else
{
    // Send email if no occupants
    sendmail
    [
        from :zoho.loginuserid
        to :"security@company.com"
        subject :"Daily Occupant Summary - " + zoho.currentdate.toString()
        message :"<h2>Daily Occupant Summary</h2><p><strong>Date:</strong> " +
zoho.currentdate.toString() + "</p><p><strong>Total Occupants:</strong> 0</p><p>No
occupants currently inside the building.</p>"
        content type :HTML
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application link

https://creatorapp.zoho.in/akash.m.2024.cse_rajalaks367/visitor-entry-tracking-system in a web browser, accept the invitation e-mail sent to your address (first-time users), sign in with your Zoho account and open Visitor Entry Tracking System. On a mobile device, install the Zoho Creator app, sign in with the same account and open the application.

D.2 Instructions for the Receptionist

- Log in with the Receptionist account.
- Open Visitor Registration. The Visitor ID is filled automatically and cannot be edited.
- Enter the visitor's name, mobile number, purpose of visit, person to meet, department, entry date and time, and set Status to "Inside"; then submit. The department head is e-mailed automatically.
- Open Visitor Log and add an entry (Action Taken = Check In), selecting the visitor by Visitor ID.
- When the visitor leaves, open All Visitor Registrations, edit the record, enter the exit date and time (it must not be earlier than the entry time) and set Status to "Checked Out"; then add a Visitor Log entry with Action Taken = Checkout.
- Use the Visitor Registrations Board to see visitors grouped by purpose of visit.

D.3 Instructions for the Security Officer

- Log in with the Security Officer account.
- Open All Visitor Registrations and filter on Status = Inside to see who is currently on the premises, or read the daily occupant summary e-mail.
- Add Visitor Log entries for check-ins, checkouts and updates; open the DashBoard for counts and charts.

D.4 Instructions for the Department Head

- Log in with the Department Head account.
- Open Department Master and keep the department name, head of department, contact e-mail, phone and Is Active flag up to date — the entry e-mail is sent to the contact e-mail.
- View registrations, logs and history through their reports (view-only).

D.5 Instructions for the System Administrator

- Add users and assign them a permission profile through Manage → Users, and maintain the role hierarchy under Settings → User Permissions.
- Check that department names in Department Master exactly match the Department values in the registration form, and that the daily schedule is active with the correct recipient.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	AKASH M	**Register No.**	240701025
Project Title	Visitor Entry Tracking System using Zoho Creator	**Date**	10.11.2026

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner:

Name: _____

Signature of External Examiner:

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	AKASH M	**Register No.**	240701025
Department / Section	Computer Science and Engineering	**Date of Viva**	10.11.2026
Application URL Verified	Yes / No	**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

Name: _____ **Name:** _____